

DATA ANALYSIS REPORT

Factors Influencing Substance Use Among Respondents

Executive Summary

This report presents the findings of a statistical analysis conducted to examine the relationship between substance use and selected social and psychological factors among respondents. The analysis focused on three major independent variables:

- Peer Pressure
- Family Members' Substance Use
- Mental Health Risk

A total of **196 valid responses** were analyzed using **Chi-Square Tests of Independence** through SPSS Crosstab Analysis. The findings revealed statistically significant relationships between all three factors and substance use behavior.

The results suggest that peer influence, family exposure to substance use, and mental health conditions play important roles in predicting or influencing substance use among individuals.

1. Introduction

Substance use remains a major public health and social concern globally. Several studies have shown that social influence, environmental exposure, and mental health conditions contribute significantly to substance use behaviors.

This analysis investigates whether:

1. Peer pressure is associated with substance use.
2. Family members' use of substances is associated with substance use.
3. Mental health risk is associated with substance use.

The study applies inferential statistical analysis using the Chi-Square Test to determine whether significant relationships exist between these variables.

2. Research Objectives

The major objectives of this analysis are:

- To determine whether peer pressure influences substance use.
 - To assess whether family members' substance use affects respondents' substance use behavior.
 - To examine the relationship between mental health risk and substance use.
-

3. Methodology

3.1 Research Design

The study adopted a quantitative research design using survey data collected from respondents.

3.2 Sample Size

A total of **196 valid cases** were analyzed.

3.3 Statistical Technique

The analysis employed:

- Crosstabulation Analysis
- Pearson Chi-Square Test of Independence

The Chi-Square test was used to determine whether there is a statistically significant relationship between the independent variables and substance use.

Decision Rule

- If **p-value < 0.05**, reject the null hypothesis.
 - If **p-value > 0.05**, fail to reject the null hypothesis.
-

4. Data Analysis and Findings

4.1 Peer Pressure and Substance Use

Crosstab Summary

Peer Pressure	Substance Use: Yes	Substance Use: No	Total
Yes	80	24	104
No	40	0	40
Not Sure	4	48	52
Total	124	72	196

Key Findings

- Among respondents who reported experiencing peer pressure, **76.9%** engaged in substance use.
- Respondents who were unsure about peer pressure showed a significantly lower rate of substance use.
- The Pearson Chi-Square value was **100.675** with a significance level of **p = 0.000**.

Interpretation

Since the p-value is less than 0.05, there is a statistically significant relationship between peer pressure and substance use.

This indicates that peer influence is a major contributing factor to substance use among respondents.

Hypothesis Decision

- **Null Hypothesis (H_0):** There is no significant relationship between peer pressure and substance use.
 - **Decision:** Reject H_0 .
-

4.2 Family Members' Substance Use and Respondents' Substance Use

Crosstab Summary

Family Members Use	Substance Use: Yes	Substance Use: No	Total
Yes	46	0	46
No	64	24	88
Not Sure	14	48	62
Total	124	72	196

Key Findings

- All respondents who indicated that family members use substances also reported substance use.
- Respondents who were unsure about family members' substance use recorded lower substance use participation.
- The Pearson Chi-Square value was **74.258** with a significance level of **p = 0.000**.

Interpretation

The result shows a statistically significant relationship between family members' substance use and respondents' substance use behavior.

This suggests that family environment and exposure significantly influence substance use habits.

Hypothesis Decision

- **Null Hypothesis (H_0):** There is no significant relationship between family members' substance use and respondents' substance use.
- **Decision:** Reject H_0 .

4.3 Mental Health Risk and Substance Use

Crosstab Summary

Mental Health Risk	Substance Use: Yes	Substance Use: No	Total
High	86	24	110

Mental Health Risk	Substance Use: Yes	Substance Use: No	Total
Low	38	48	86
Total	124	72	196

Key Findings

- Respondents with high mental health risk showed higher substance use prevalence.
- Approximately **78.2%** of respondents with high mental health risk engaged in substance use.
- The Pearson Chi-Square value was **24.002** with a significance level of **p = 0.000**.

Interpretation

There is a statistically significant relationship between mental health risk and substance use.

This finding indicates that individuals with higher mental health risk are more likely to engage in substance use.

Hypothesis Decision

- **Null Hypothesis (H_0):** There is no significant relationship between mental health risk and substance use.
- **Decision:** Reject H_0 .

5. Overall Discussion

The analysis revealed that all three independent variables examined in this study have statistically significant relationships with substance use.

Major Observations

- Peer pressure strongly influences substance use behavior.
- Exposure to substance use within the family environment increases the likelihood of substance use.
- Poor mental health conditions are associated with higher substance use rates.

These findings align with existing literature which suggests that substance use is influenced by both social and psychological factors.

6. Conclusion

The study concludes that:

- Peer pressure significantly affects substance use.
- Family members' substance use significantly influences respondents' substance use behavior.
- Mental health risk significantly contributes to substance use.

All null hypotheses tested in this analysis were rejected because the significance values obtained were below the 0.05 threshold.

The findings emphasize the need for interventions that target peer influence, family environments, and mental health support systems in addressing substance use.

7. Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. **Peer Education Programs:** Educational institutions and communities should implement peer mentorship and awareness programs to reduce negative peer influence.
2. **Family-Based Interventions:** Families should be educated on the impact of substance use within the home environment.
3. **Mental Health Support Services:** Increased access to counseling and mental health services should be provided to individuals at risk.
4. **Public Awareness Campaigns:** Government agencies and NGOs should intensify campaigns on the dangers of substance use.
5. **Early Detection and Prevention:** Schools and healthcare institutions should adopt early screening strategies for substance use risk factors.

8. Statistical Summary Table

Variable Tested	Chi-Square Value	Degrees of Freedom	P-value	Decision
Peer Pressure vs Substance Use	100.675	2	0.000	Significant
Family Members Use vs Substance Use	74.258	2	0.000	Significant

Variable Tested	Chi-Square Value	Degrees of Freedom	p-value	Decision
Mental Health Risk vs Substance Use	24.002	1	0.000	Significant

9. Appendix

Statistical Technique Used

Pearson Chi-Square Test

The Pearson Chi-Square Test of Independence is used to determine whether there is a significant association between two categorical variables.

The test compares observed frequencies with expected frequencies to identify whether differences are statistically significant.

Significance Level

The study used a 95% confidence level with a significance threshold of:

- $\alpha = 0.05$

Any p-value below 0.05 indicates a statistically significant relationship.

Prepared Data Analysis Report

Software Used: SPSS

Total Respondents: 196

Analysis Type: Crosstabulation and Chi-Square Test of Independence